4. Confounding and the Back-door Criterion

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Confounding and Backdoor* (Prof. Sanghack Lee, GSDS SNU)

The central question is how a causal effect can be identified when some parents of the treatment are unobserved, and how the resulting adjustment formula can be evaluated from finite samples.

These notes consolidate repeated slides, complete the proof that back-door adjustment is equivalent to adjustment by direct parents, verify the non-identifiability example numerically, and derive the IPW estimator step by step. Physical PDF page numbers are listed in Section 6.

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1 Recap: identifiable and non-identifiable effects

Purpose. Illustrate both sides of the identifiability definition with concrete pairs of models.

1.1 An identifiable effect

Take $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$ (no latent confounding). Any compatible model factorises as

$$\begin{aligned} P(\mathbf{v}) &= P(z)P(x \mid z)P(y \mid x, z), \\ P(\mathbf{v}' \mid \mathrm{do}(x)) &= P(z)P(y \mid x, z), \\ P(y \mid \mathrm{do}(x)) &= \sum_z P(z)P(y \mid x, z). \end{aligned}$$

Consider two models

$$\mathcal{M}^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow f_x^{(j)}(z, u_x) \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$\mathcal{M}_x^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow x \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$j = 1, 2.$$

Whatever the functions and $P(\mathbf{u})$, if $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ agree on $\langle \mathcal{G}, P(\mathbf{v}) \rangle$ they agree on $P(z)$ and $P(y \mid x, z)$, hence on $P(\mathbf{v}' \mid \mathrm{do}(x))$.

1.2 A non-identifiable effect

Take $\mathcal{G}$: $X \to Y$ and $X \leftrightarrow Y$ (through a latent U_{xy}). Let U_y, U_{xy} be independent fair coins in both models and $X \leftarrow U_{xy}$ in the observational world. Define

$$\mathcal{M}^{(1)} : \; Y \leftarrow (X \oplus U_{xy}) \vee U_y, \qquad \mathcal{M}^{(2)} : \; Y \leftarrow U_y.$$

Observationally $X = U_{xy}$, so $X \oplus U_{xy} = 0$ and $Y = U_y$ in both models: the observational distributions coincide. Under $\mathrm{do}(x)$:

U_y	U_{xy}	X	$Y^{(1)}$	$Y^{(2)}$	$P(\mathbf{u})$
0	0	x	x	0	1/4
0	1	x	$\neg x$	0	1/4
1	0	x	1	1	1/4
1	1	x	1	1	1/4

Y	$P^{(1)}(Y \mid \mathrm{do}(x))$	$P^{(2)}(Y \mid \mathrm{do}(x))$
0	1/4	1/2
1	3/4	1/2

In $\mathcal{M}^{(1)}$, $Y = 0$ requires $U_y = 0$ and $x = U_{xy}$, probability 1/4. Both models induce $\mathcal{G}$ and the same $P(\mathbf{v})$, yet $P^{(1)}(y \mid \mathrm{do}(x)) \neq P^{(2)}(y \mid \mathrm{do}(x))$. In general $P(\mathbf{v} \mid \mathrm{do}(x)) = \sum_{u_{xy}} P(y \mid x, u_{xy})P(u_{xy})$ depends on the latent U_{xy}, which the data do not reveal.

2 Confounding bias

Purpose. Show how a common cause makes conditioning differ from intervening.

What is the causal effect of exercise on cholesterol? Age affects both (age $\to$ exercise, age $\to$ cholesterol, exercise $\to$ cholesterol). In the scatter plot, older people exercise more and also have higher cholesterol, so pooled data show cholesterol *increasing* with exercise, while within every age group more exercise means lower cholesterol. Hence

$$P(\text{cholesterol} \mid \text{exercise}) \neq P(\text{cholesterol} \mid \text{do(exercise)}).$$

This difference is **confounding bias**, one of the major obstacles to causal inference and interpretability.

Is confounding bias removable? Goal: find $Q = P(y \mid \text{do}(x))$ from measurements on $Z_1, \ldots, Z_k$ when some parents of X are unobserved. Adjustment by direct parents (Lecture 3) is then unavailable. Which subsets of measured variables can replace the parents?

3 The back-door criterion

Purpose. Give a graphical condition under which a covariate set can substitute for the parents of the treatment.

Definition 3.1 (Back-door criterion; Pearl 2000, Def. 3.3.1)**.** A set $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if

(i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$; and

(ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ that contains an arrow into X (a *back-door path*).

Theorem 3.2 (Back-door adjustment)**.** *If $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$, the effect is identifiable and*

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

3.1 Back-door sets substitute for the parents: an example

In the sprinkler network, Rain satisfies the criterion relative to (Sprinkler, Wet): (i) Rain is not a descendant of Sprinkler; (ii) Rain blocks the only back-door path Sprinkler $\leftarrow$ Season $\to$ Rain $\to$ Wet. Starting from adjustment for the direct parent Season,

$$\begin{aligned}
P(\text{wt} \mid \text{do(sp)}) &= \sum_{\text{sn}} P(\text{wt} \mid \text{sp}, \text{sn}) P(\text{sn}) \\
&= \sum_{\text{sn,rn}} P(\text{wt} \mid \text{sp}, \text{sn}, \text{rn}) P(\text{rn} \mid \text{sp}, \text{sn}) P(\text{sn}) \\
&= \sum_{\text{sn,rn}} P(\text{wt} \mid \text{sp}, \text{rn}) P(\text{rn} \mid \text{sn}) P(\text{sn}) \\
&= \sum_{\text{rn}} P(\text{wt} \mid \text{sp}, \text{rn}) \sum_{\text{sn}} P(\text{rn}, \text{sn}) \\
&= \sum_{\text{rn}} P(\text{wt} \mid \text{sp}, \text{rn}) P(\text{rn}).
\end{aligned}$$

The third equality uses Wet $\perp\!\!\!\perp$ Season | Sprinkler, Rain and Rain $\perp\!\!\!\perp$ Sprinkler | Season, both read off by d-separation.

3.2 General proof: back-door adjustment = parent adjustment

Let $Pa^-_{\mathbf{X}}$ be the parents of $\mathbf{X}$ outside $\mathbf{X}$, and $\mathbf{Z}^- = \mathbf{Z} \setminus Pa^-_{\mathbf{X}}$. The two conditions imply two independences:

$$\begin{aligned} &\text{(i)} \implies \mathbf{X} \perp\!\!\!\perp (\mathbf{Z} \setminus Pa^-_{\mathbf{X}}) \mid Pa^-_{\mathbf{X}} && \text{(local Markov)},\\ &\text{(ii)} \implies \mathbf{Y} \perp\!\!\!\perp (Pa^-_{\mathbf{X}} \setminus \mathbf{Z}) \mid \mathbf{Z}, \mathbf{X} && \text{(back-door paths blocked)}. \end{aligned}$$

The first holds because $\mathbf{Z}$ consists of non-descendants of $\mathbf{X}$; the second because every path from a parent of $\mathbf{X}$ to $\mathbf{Y}$ that avoids $\mathbf{X}$ extends to a back-door path. Then

$$\begin{aligned} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) &= \sum_{\mathbf{pa}^-_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}^-_{\mathbf{X}}) P(\mathbf{pa}^-_{\mathbf{X}}) && \text{parent adjustment}\\ &= \sum_{\mathbf{z}^-, \mathbf{pa}^-_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}^-_{\mathbf{X}}, \mathbf{z}^-) P(\mathbf{z}^- \mid \mathbf{x}, \mathbf{pa}^-_{\mathbf{X}}) P(\mathbf{pa}^-_{\mathbf{X}}) && \text{def. of cond. probability}\\ &= \sum_{\mathbf{z}^-, \mathbf{pa}^-_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^- \mid \mathbf{pa}^-_{\mathbf{X}}) P(\mathbf{pa}^-_{\mathbf{X}}) && \text{(ii) and (i)}\\ &= \sum_{\mathbf{z}^-, \mathbf{pa}^-_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^-, \mathbf{pa}^-_{\mathbf{X}})\\ &= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) \sum_{\mathbf{pa}^-_{\mathbf{X}} \setminus \mathbf{z}} P(\mathbf{z}^-, \mathbf{pa}^-_{\mathbf{X}}) && \text{by algebra}\\ &= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}). \end{aligned}$$

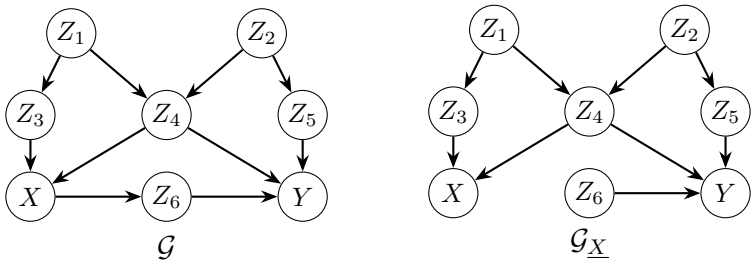
In the third line, $\{\mathbf{x}, \mathbf{pa}^-_{\mathbf{X}}, \mathbf{z}^-\} = \{\mathbf{x}, \mathbf{z}\} \cup (\mathbf{pa}^-_{\mathbf{X}} \setminus \mathbf{z})$, so (ii) removes $\mathbf{pa}^-_{\mathbf{X}} \setminus \mathbf{z}$ from the conditioning set; (i) removes $\mathbf{x}$ from $P(\mathbf{z}^- \mid \cdot)$. *Adjustment by* $\mathbf{Z}$ *is equivalent to adjustment by the direct parents whenever* $\mathbf{Z}$ *is back-door admissible.* (The parent formula holds in the Markovian model containing all variables, latent ones included; the final expression uses only measured variables.)

3.3 A graphical test

Condition (ii) says that $\mathbf{Z}$ d-separates $\mathbf{X}$ from $\mathbf{Y}$ once the causal paths are removed. Let $\mathcal{G}_{\underline{\mathbf{X}}}$ be $\mathcal{G}$ with all edges *out of* $\mathbf{X}$ deleted. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) \text{ is identified by } \mathbf{Z} \quad \text{if (i) and } (\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}.$$

Example 3.3 (Pearl's seven-variable model)**.** $Z_1 \to Z_3 \to X$, $Z_1 \to Z_4$, $Z_2 \to Z_4$, $Z_2 \to Z_5 \to Y$, $Z_4 \to X$, $Z_4 \to Y$, $X \to Z_6 \to Y$.



$\mathcal{G}$ $\qquad$ $\mathcal{G}_{\underline{X}}$

The back-door paths are $X \leftarrow Z_4 \to Y$, $X \leftarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \to Y$ and $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \leftarrow Z_2 \to Z_5 \to Y$. Conditioning on Z_4 blocks the first three but opens the collider Z_4 on the last, which Z_1 blocks again. Hence $\{Z_1, Z_4\}$ is admissible:

$$P(y \mid \mathrm{do}(x)) = \sum_{z_1, z_4} P(y \mid x, z_1, z_4) P(z_1, z_4).$$

Other admissible sets include $\{Z_3, Z_4\}$, $\{Z_4, Z_2\}$, $\{Z_4, Z_5\}$; $\{Z_4\}$ alone is not admissible and Z_6 never is (descendant of X).

Example 3.4 (More back-door exercises)**.** (a) $Z_2 \to Z_4$, $Z_2 \to Z_5$, $Z_4 \to X$, $Z_4 \to Y$, $X \leftrightarrow Z_4$, $X \to Z_6 \to Y$, $Z_5 \to Y$. Z_4 must be included to block $X \leftarrow Z_4 \to Y$; but Z_4 is a collider on $X \leftrightarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, so Z_2 or Z_5 must be added. Admissible: $\mathbf{Z} \in \{\{Z_4, Z_2\}, \{Z_4, Z_5\}, \{Z_4, Z_2, Z_5\}\}$.
(b) $X \to Z$, $X \to Y$. There is no back-door path and Z is a descendant of X, so $\mathbf{Z} = \emptyset$ and $P(y \mid \mathrm{do}(x)) = P(y \mid x)$.

4 Evaluating back-door adjustment: inverse probability weighting

Purpose. Turn the adjustment formula into an estimator whose cost does not explode with the dimension of $\mathbf{Z}$.

The criterion says when $\mathbf{Z}$ is admissible,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}),$$
$$\mathbb{E}[Y \mid \mathrm{do}(\mathbf{x})] = \mathbb{E}_{\mathbf{Z}}\big[\mathbb{E}[Y \mid \mathbf{x}, \mathbf{Z}]\big].$$

Evaluating it naively requires estimating many conditional distributions and summing over a possibly high-dimensional $\mathbf{Z}$, roughly $O(\exp(|\mathbf{Z}|))$ time, with sparse cells for each $\mathbf{z}$.

4.1 Rewriting with the propensity score

$$P(y \mid \mathrm{do}(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z}) P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x, \mathbf{z})} P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})}.$$

The numerator is an entry of the joint distribution; the denominator is the assignment probability (propensity), fitted by a model $g(\mathbf{z}) \approx P(x \mid \mathbf{z})$.

4.2 Sample version

With N samples, replace $P(y, x, \mathbf{z})$ by its empirical frequency:

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &\approx \sum_{\mathbf{z}} \frac{\frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \sum_{\mathbf{z}} \frac{\mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \frac{\mathbb{1}(y_i = y, x_i = x)}{g(\mathbf{z}_i)}. \end{aligned}$$

The sum over $\mathbf{z}$ collapses because only $\mathbf{z} = \mathbf{z}_i$ contributes. The cost is proportional to N, not to $|\mathfrak{X}_\mathbf{Z}|$. ($\mathbb{1}$(predicate) is 1 if the predicate is true and 0 otherwise.) Similarly,

$$\begin{aligned}\mathbb{E}[Y \mid \mathrm{do}(x)] &= \sum_{\mathbf{z},y} y \, \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})} \\ &\approx \frac{1}{N} \sum_{i=1}^{N} \sum_{\mathbf{z},y} \frac{y \, \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \frac{y_i \, \mathbb{1}(x_i = x)}{g(\mathbf{z}_i)}.\end{aligned}$$

4.3 Average treatment effect

For binary X with $g(\mathbf{z}) = P(X = 1 \mid \mathbf{z})$, when the back-door criterion holds,

$$\begin{aligned}\widehat{\mathrm{ATE}} &= \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 1)] - \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 0)] \\ &= \frac{1}{N} \sum_{i=1}^{N} y_i \left(\frac{\mathbb{1}(x_i = 1)}{g(\mathbf{z}_i)} - \frac{\mathbb{1}(x_i = 0)}{1 - g(\mathbf{z}_i)} \right) \\ &= \frac{1}{N} \sum_{i=1}^{N} \left(\frac{y_i x_i}{g(\mathbf{z}_i)} - \frac{y_i (1 - x_i)}{1 - g(\mathbf{z}_i)} \right).\end{aligned}$$

Notation varies across fields: covariates are often X or L, treatment T, W or A.

4.4 Interpretation: pseudo-causal samples

IPW turns observational samples into a "pseudo-population" that mimics a randomised experiment: samples with $x_i \neq x$ are discarded and each remaining sample is replicated $1/g(\mathbf{z}_i)$ times. A unit with $P(x_1 \mid \mathbf{z}_1) = 1/3$ counts three times, one with $P(x_3 \mid \mathbf{z}_3) = 1/2$ counts twice. Units that were unlikely to receive treatment given their covariates stand in for the many similar units that were not treated, removing the dependence between X and $\mathbf{Z}$.

5 Summary

- Confounding bias arises when treatment and outcome share a common cause, making $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$.
- The back-door criterion is a graphical test of whether $\mathbf{Z}$ is admissible for adjustment.
- Back-door adjustment: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_\mathbf{z} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.
- It generalises adjustment by direct parents: any back-door admissible set can substitute for the parents.
- IPW evaluates the back-door expression from samples without summing over a high-dimensional $\mathbf{Z}$.

6 Source coverage

The source deck has 39 physical pages (20 numbered slides with build-ups).

Topic	Physical pages
Recap: identifiable and non-identifiable examples	1–7
Confounding bias, exercise–cholesterol example	8–15
Back-door criterion, theorem, sprinkler example, proof	16–22
Graphical test and exercises	23–27
Evaluation, IPW, ATE, pseudo-samples, summary	28–39